2. Client-Side Parsing & Standards Compliance

Traditional web architectures rely on server-side microservices to convert and display e-books. However, modern browser runtimes possess powerful native capabilities:

Standard EPUB Package Anatomy

An EPUB is fundamentally an unencrypted ZIP archive containing:

1. META-INF/container.xml pointing to the root Open Packaging Format (OPF) file.
2. The OPF file which declares the metadata, manifest items, and reading spine.
3. Clean XHTML documents styled with standard CSS.
4. Embedded images and fonts.

The Zero-Overhead In-Browser Reader

By utilizing the browser native DecompressionStream and Uint8Array buffers, client devices can parse chapters in milliseconds directly in memory. This eliminates server storage quotas, preserves privacy, and allows instant reading even when completely disconnected from the network.

3. Building a Personal Digital Library Shelf

Personal archives must endure across years and system migrations.
To make archiving frictionless for everyone:

1. Layman Workflow: Simply drop any .epub or .pdf into books/archive/.
2. Automatic Metadata Harvesting: The indexer automatically extracts titles, authors, descriptions, and chapter counts without requiring manual configuration files.
3. Dual Consumption Modes: Visitors can download files directly to e-ink hardware (Kindle, Kobo, reMarkable) or read instantly in the built-in browser reader.

Conclusion & Next Steps

With this setup, your site doubles as an interactive publication and a private digital vault.
Your library remains entirely your own: portable, lightweight, and sovereign.